

A Formal Mathematical Theory of Triadic Structural Closure

Support, Structure, Linkage, Coherence, Unity and Fragility in the PETU Framework

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Claim status and scope

Document type	Formal mathematical companion to the empirical PETU atlas.
Claim level	Candidate cross-domain structural principle.
Primary object	Role-equivalent closure architecture among support (H), structure (S), linkage (L) and closure (Φ).
Not claimed	Not a completed universal law, not a theory of everything, not a replacement for domain sciences, not a theological proof, and not yet confirmed by fully preregistered independent replication.
Empirical relation	Companion formalism to the public preprint and reproducibility package.

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Abstract

This document develops a formal mathematical companion to the empirical PETU atlas. It formalizes the hypothesis that coherent natural systems may often be described by a role-equivalent architecture involving support (H), structure (S) and linkage (L), with system-level coherence or functional closure denoted by Φ . The theory does not assume that different natural domains share identical mechanisms. Instead, it defines a common structural grammar: support becomes organized into structure, structure enables linkage, and linkage contributes to closure at the system level. The document introduces normalized triadic states, admissible systems, unity, balance, dominance, fragility, co-inherence, triadic gain, non-circularity requirements, perturbative response, multi-scale closure, categorical comparison, tensorial extension, variational stability and law-candidate criteria. It also formalizes the distinction between raw support H_{raw} and structured support $H_{\text{structured}}$, which is necessary for interpreting apparent support-only effects.

1. Notation table

Symbol	Meaning
X	Natural system under analysis.
Ω_X	Observational domain of X : time windows, regions, nodes, residues, cells, sites, basins, grid cells, galaxies or comparable units.
ω	Observational unit or state, $\omega \in \Omega_X$.

H_X	Support function: substrate, capacity, resource, energy, abundance, amplitude, flow or enabling magnitude.
S_X	Structure function: constraint, topology, modularity, spatial/temporal form, hierarchy or organization.
L_X	Linkage function: coupling, interaction, synchrony, binding, transfer, connectivity, co-regulation or relational integration.
Φ_X	Closure/coherence variable: system-level coherence, unity, integration, stability, resilience or functional closure.
T_X	Triadic state (H_X, S_X, L_X) .
U_X	Unity operator.
B_X	Balance/proportionality operator.
D_X^{gap}, D_X^{share}	Dominance operators.
F_X	Fragility operator.
R_X	Rupture, instability or discontinuity measure.
G_3	Irreducible triadic gain over monadic and dyadic baselines.
π_i	Perturbation operator targeting component $i \in \{H, S, L\}$.
$\mathcal{B}$	Baseline model set.
$\mathcal{N}^*$	Class of admissible natural systems for which H, S, L, Φ are meaningfully measurable without strong circularity.

2. Admissible systems and observation domains

Let X be a natural system and Ω_X its observational domain. A measured state is indexed by $\omega \in \Omega_X$. A PETU-admissible system belongs to a class $\mathcal{N}^*$ if there exist declared, measurable and normalized functions

$$H_X, S_X, L_X, \Phi_X : \Omega_X \rightarrow [0, 1],$$

such that H_X, S_X and L_X preserve their structural roles and Φ_X is either external, derived with declared construction, or explicitly labelled as an internal closure endpoint.

The triadic state is

$$T_X(\omega) = (H_X(\omega), S_X(\omega), L_X(\omega)).$$

For dynamic systems, ω may include time:

$$\omega = (i, t),$$

where i denotes a unit and t denotes time.

3. Structural-role functions

3.1. Support

Support is the enabling substrate or capacity of the system:

$$H_X = \text{support function of } X.$$

Depending on the domain, it may represent energy, mass, abundance, signal capacity, flow magnitude, resource availability, density, amplitude, reaction capacity or substrate potential.

3.2. Structure

Structure is support under constraint:

$$S_X = \text{structure function of } X.$$

It may represent topology, modularity, shape, hierarchy, order, spatial organization, temporal recurrence, folding, pattern, network form or internal constraint.

3.3. Linkage

Linkage is structure in relation:

$$L_X = \text{linkage function of } X.$$

It may represent coupling, interaction, synchrony, transfer, binding, flow, connectivity, relational continuity, teleconnection, co-regulation or integration.

3.4. Closure

Closure is the degree to which support, structure and linkage cohere as an organized whole:

$$\Phi_X = \text{closure function of } X.$$

Depending on the domain, Φ_X may represent coherence, unity, resilience, integration, functional stability, system-level order, global organization or natural-state integrity.

4. Raw support and structured support

A central methodological distinction is

$$H_{\text{raw}} \neq H_{\text{structured}}.$$

Raw support is minimally organized support:

$$H_{\text{raw}} = \text{support prior to explicit structural inscription.}$$

Structured support is support already containing structural information:

$$H_{\text{structured}} = H_{\text{raw}} \oplus S,$$

where $\oplus$ denotes a domain-specific operation of structural inscription.

Therefore, a model

$$H \rightarrow L$$

does not necessarily imply that pure support generates linkage. If $H = H_{\text{structured}}$, then

$$H \rightarrow L \quad \text{may mean} \quad (H_{\text{raw}} \oplus S) \rightarrow L.$$

The clean emergence test is therefore

$$H_{\text{raw}} + S \rightarrow L \quad \text{versus} \quad H_{\text{raw}} \rightarrow L.$$

A high-performance $H \rightarrow L$ result is evidence for monadic support only if H has been shown to be genuinely raw.

5. Mathematical assumptions

The formal theory assumes:

1. $H_X, S_X, L_X, \Phi_X \in [0, 1]$ after declared normalization.
2. The variables are role-equivalent, not physically identical across domains.
3. The normalization method, observation units, grouping structure and scale parameters are declared.
4. Φ_X is protected against circularity by endpoint labelling and anti-leakage checks.
5. Performance functionals P are declared before confirmatory testing.
6. Group-aware validation is required when observations are nested by subject, site, region, model, scale or time.
7. Perturbations are auxiliary stress-tests unless the perturbation is independently justified as domain-relevant.

6. Unity, balance, dominance and fragility

6.1. Unity

The minimal unity operator is the geometric mean:

$$U_X = (H_X S_X L_X)^{1/3}.$$

If any essential component collapses, unity collapses:

$$H_X = 0 \Rightarrow U_X = 0, \quad S_X = 0 \Rightarrow U_X = 0, \quad L_X = 0 \Rightarrow U_X = 0.$$

Unity is therefore conjunctive rather than additive.

6.2. Balance

Because $H_X, S_X, L_X \in [0, 1]$, define

$$B_X = 1 - (\max\{H_X, S_X, L_X\} - \min\{H_X, S_X, L_X\}).$$

Then $B_X \in [0, 1]$. If the three components are equal, $B_X = 1$. If one component is maximal and another minimal, B_X approaches 0.

6.3. Dominance

Dominance must be defined unambiguously. Two useful variants are:

$$D_X^{gap} = \max\{H_X, S_X, L_X\} - \text{median}\{H_X, S_X, L_X\},$$

and

$$D_X^{share} = \frac{\max\{H_X, S_X, L_X\}}{H_X + S_X + L_X + \varepsilon}, \quad \varepsilon > 0.$$

D_X^{gap} measures excess over the median, while D_X^{share} measures capture of the total triadic mass.

6.4. Fragility

A minimal fragility operator is

$$F_X = 1 - U_X B_X.$$

A richer form is

$$F_X = \alpha(1 - U_X) + \beta(1 - B_X) + \gamma D_X^{gap} + \delta R_X, \quad \alpha, \beta, \gamma, \delta \geq 0.$$

Here R_X denotes rupture, discontinuity or instability. Fragility is relational: it is not merely the absence of one component, but the failure of coherent relation among the components.

7. Closure functions and triadic gain

The general closure relation is

$$\Phi_X \approx \mathcal{F}_X(H_X, S_X, L_X).$$

A minimal interpretable statistical model is

$$\Phi_X = \beta_0 + \beta_H H_X + \beta_S S_X + \beta_L L_X + \beta_{HS} H_X S_X + \beta_{HL} H_X L_X + \beta_{SL} S_X L_X + \beta_{HSL} H_X S_X L_X + \varepsilon_X.$$

The triadic interaction term is $\beta_{HSL} H_X S_X L_X$. However, PETU does not require the true closure function to be polynomial. $\mathcal{F}_X$ may be nonlinear, dynamical, graph-based, information-theoretic or topological.

Let

$$\mathcal{M}_1 = \{H, S, L\}, \quad \mathcal{M}_2 = \{H + S, H + L, S + L\}, \quad \mathcal{M}_3 = \{H + S + L\}.$$

Let $P(\mathcal{M})$ be a declared performance functional. The primary difference-form triadic gain is

$$G_3^{diff} = P(\mathcal{M}_3) - \max_{\mathcal{M} \in \mathcal{M}_1 \cup \mathcal{M}_2} P(\mathcal{M}).$$

A ratio form may also be reported:

$$G_3^{ratio} = \frac{P(\mathcal{M}_3)}{\max_{\mathcal{M} \in \mathcal{M}_1 \cup \mathcal{M}_2} P(\mathcal{M}) + \varepsilon}.$$

Triadic gain is strong when

$$G_3^{diff} > \tau_G$$

for a preregistered threshold τ_G .

8. Non-circularity and endpoint independence

The main methodological risk is circular closure: constructing Φ_X directly from H_X, S_X, L_X and then reporting that H, S, L reconstruct Φ . PETU therefore distinguishes:

1. **Internal closure reconstruction:** Φ_X is derived from structural features and must be labelled internal.
2. **Derived unity:** $U_X = (H_X S_X L_X)^{1/3}$ is a defined PETU index, not an independent endpoint.
3. **External endpoint prediction:** an independent endpoint Y_X is not used in constructing H, S, L, Φ .

4. **Fragility prediction:** F_X is derived or external, and its construction must be stated.
5. **Natural-versus-perturbed discrimination:** class labels must not leak into feature construction.
6. **Anti-leakage validation:** grouped cross-validation, held-out domains, held-out sites or held-out time blocks are required where applicable.

A strong confirmatory test should include at least one endpoint that is external to the construction of H, S, L .

9. Perturbations, null models and falsification

Let π_H, π_S, π_L be perturbation operators targeting support, structure or linkage. For relevant disruptive perturbations, PETU predicts

$$\Phi_X > \Phi_{\pi_i(X)} \quad \text{or} \quad F_{\pi_i(X)} > F_X, \quad i \in \{H, S, L\}.$$

The perturbative response is

$$\Delta_\Phi^i = \Phi_X - \Phi_{\pi_i(X)}.$$

Null models may include label shuffling, feature shuffling, within-group permutation, block shuffling, degree-preserving graph randomization, phase randomization, spatially constrained nulls and matched-complexity random features.

The theory is weakened or falsified when:

1. independent $HSL \rightarrow \Phi$ fails systematically;
2. HSL does not outperform H -only when H_{raw} is properly isolated;
3. perturbations fail to reduce Φ or increase fragility in predicted contexts;
4. alternative operationalizations erase the pattern;
5. cross-domain convergence depends on circular Φ definitions;
6. preregistered replication fails in anchor domains;
7. external endpoints fail under anti-leakage validation.

10. Multi-scale and dynamical closure

Let $r \in \mathcal{R}_X$ be an observational scale. A scale-indexed triadic system is

$$H_X(r), S_X(r), L_X(r), \Phi_X(r).$$

Multi-scale closure occurs when

$$\Phi_X(r) \approx \mathcal{F}_{X,r}(H_X(r), S_X(r), L_X(r))$$

across a declared set of scales.

For dynamic systems:

$$H = H(t), \quad S = S(t), \quad L = L(t), \quad \Phi = \Phi(t).$$

A general dynamical form is

$$\begin{aligned}\frac{dH}{dt} &= f_H(H, S, L, \theta_H), & \frac{dS}{dt} &= f_S(H, S, L, \theta_S), \\ \frac{dL}{dt} &= f_L(H, S, L, \theta_L), & \frac{d\Phi}{dt} &= f_\Phi(H, S, L, \Phi, \theta_\Phi).\end{aligned}$$

Triadic dynamical closure predicts

$$\Delta\Phi \approx \mathcal{D}(\Delta H, \Delta S, \Delta L).$$

11. Co-inherence and non-separability

The relation among H, S, L can be represented by a co-inherence matrix:

$$C_X = \begin{pmatrix} 1 & C_{HS} & C_{HL} \\ C_{SH} & 1 & C_{SL} \\ C_{LH} & C_{LS} & 1 \end{pmatrix}, \quad C_{ij} \in [0, 1].$$

The components are distinguishable but may not be separable. A system contains irreducible triadic information when an interaction object is non-zero:

$$I_{HSL}(X_T) \neq 0.$$

This may be operationalized by interaction regression terms, synergy measures, tensor contractions, topological invariants or model comparison.

12. Categorical structural equivalence

Let **TriSys** be the category of triadically structured systems. An object is

$$X_T = (X, H_X, S_X, L_X, \Phi_X).$$

A morphism

$$f : X_T \rightarrow Y_T$$

maps observational units through $f_\Omega : \Omega_X \rightarrow \Omega_Y$ and preserves structural roles up to admissible distortion:

$$H_Y(f_\Omega(\omega)) \approx h_f(H_X(\omega)),$$

and similarly for S, L, Φ .

A closure map is

$$\Theta_X : H_X \times S_X \times L_X \rightarrow \Phi_X.$$

The categorical layer is a formal language for structural comparison, not an independent proof of PETU. The measurable non-commutativity error is

$$E_\Theta(f) = \|f_\Phi \circ \Theta_X - \Theta_Y \circ f_{HSL}\|.$$

A morphism is approximately closure-preserving when

$$E_\Theta(f) < \tau_\Theta.$$

13. Tensorial and field-theoretic extension

For high-dimensional systems, scalar components can be generalized:

$$H_i, \quad S_{ij}, \quad L_{ijk}.$$

A tensorial closure form is

$$\Phi_X = \mathcal{C}_X[H_i, S_{ij}, L_{ijk}],$$

with a minimal contraction

$$\Phi_X = \sum_{i,j,k} W_X^{ijk} H_i S_{ij} L_{ijk}.$$

In a field-theoretic direction, one may define fields over a manifold or observational space:

$$H(x), \quad S(x), \quad L(x), \quad \Phi(x),$$

and a local closure density

$$\phi(x) = \mathcal{F}_x(H(x), S(x), L(x)).$$

These extensions are mathematical research horizons; they are not yet established physical theories.

14. Variational stability and phase transitions

A variational version can be written using a triadic action:

$$\mathcal{A}_X[T] = \int (E_\Phi(T) + \lambda_U(1 - U_X) + \lambda_B(1 - B_X) + \lambda_D D_X^{gap} + \lambda_R R_X + V_T(T)) dt.$$

Coherent regimes are candidate low-action configurations:

$$\delta \mathcal{A}_X = 0.$$

Fragility rises when the triadic potential increases:

$$V_T \uparrow, \quad F_X \rightarrow F_c, \quad \Psi \downarrow.$$

A phase transition may occur when

$$\lambda_{\min}(\nabla_T^2 V_T) = 0,$$

or when a small triadic displacement produces a large coherence shift:

$$\|\Delta T\| < \varepsilon_T \quad \Rightarrow \quad |\Delta \Phi| > \tau_\Phi.$$

15. Law-candidate criteria

Triadic structural closure should be formulated as a law-candidate only over the admissible class $\mathcal{N}^*$. A system $X \in \mathcal{N}^*$ satisfies strong triadic closure when:

1. H_X, S_X, L_X, Φ_X are measurable and normalized;
2. $\Phi_X \approx \mathcal{F}_X(H_X, S_X, L_X)$;
3. $P(HSL \rightarrow \Phi) > \tau_\Phi$;
4. $G_3^{diff} > \tau_G$, or apparent reducibility is explained by the $H_{\text{raw}}/H_{\text{structured}}$ distinction;
5. perturbations of H, S, L reduce Φ or increase F ;
6. the architecture survives independent validation or preregistered replication.

This is a formal empirical criterion, not a purely deductive theorem.

16. Relation to the empirical Atlas 4–4–4

The empirical atlas provides observational support for:

$$HSL \rightarrow \Phi, \quad HSL \rightarrow U, \quad HSL \rightarrow F,$$

and for natural-versus-perturbed discrimination across micro, meso and macro systems. The independent tier-1 validations in immune, metabolic, hydrological and climatic systems motivate continued formal development. The appropriate status remains:

candidate cross-domain structural principle

not:

established universal law.

17. Limitations

The formal theory remains dependent on empirical operationalizations. Cross-domain equivalence is structural, not mechanistic. Φ must be protected against circular definition. The $H_{\text{raw}}/H_{\text{structured}}$ distinction is essential and may not always be empirically resolvable. Tensorial, categorical and variational extensions are proposed research horizons. The present evidence supports a candidate structural principle, not an established universal law.

18. Open problems

1. Define invariant transformations for H, S, L across domains.
2. Establish non-circular external Φ endpoints.
3. Prove when triadic gain cannot arise from feature redundancy.
4. Formalize co-inherence using information decomposition.
5. Build raw/structured support decompositions.
6. Derive tensorial closure from domain-specific dynamics.
7. Test preregistered predictions on untouched datasets.
8. Determine whether PETU admits a physical action principle.

19. Roadmap

Discovery atlas $\rightarrow$ independent validation $\rightarrow$ preregistered replication $\rightarrow$ mathematical refinement $\rightarrow$ law-candidate

Appendix A. Master equations

$$T_X = (H_X, S_X, L_X)$$

$$U_X = (H_X S_X L_X)^{1/3}$$

$$B_X = 1 - (\max\{H_X, S_X, L_X\} - \min\{H_X, S_X, L_X\})$$

$$D_X^{\text{gap}} = \max\{H_X, S_X, L_X\} - \text{median}\{H_X, S_X, L_X\}$$

$$F_X = \alpha(1 - U_X) + \beta(1 - B_X) + \gamma D_X^{gap} + \delta R_X$$

$$\Phi_X \approx \mathcal{F}_X(H_X, S_X, L_X)$$

$$G_3^{diff} = P(H + S + L \rightarrow \Phi) - \max_{\mathcal{B}} P(\mathcal{B})$$

$$L_X \approx \mathcal{G}_X(H_{\text{raw}}, S_X)$$

$$H_{\text{structured}} = H_{\text{raw}} \oplus S_X$$